Everybody Owns a Piece: Running the Switches

Oct 18, 2026 · @Chris

The short answer

The setup that holds up best puts the whole quarter in the citizen fund, keeps the profit share for workers, and pays the fund out of the cash it actually receives. Of the thirteen ways of handling workers' shares, three clear all four pass lines on all ten market paths under that payout rule, and this one pays the citizen fund twice what the other two do.

Switch	What holds up	Settled by
Worker exit	No worker trust. The whole quarter goes to the citizen fund; workers keep the profit share, paid to them every year (half to spend, half into their own locked account)	The numbers rule out all but three designs; this one pays everyone twice as much. Whether workers should also own a piece is yours
Payout rule	80% of the fund's share of the cash companies paid owners, averaged over five years. Never more than cash in plus the reserve; reserve capped at two years of checks	The numbers, inside a close range that's yours
Lock	Carbon checks fully spendable, half of everything else locked	The carbon part leans on the numbers; the level is yours
Carbon	In, with the full fee sent back and the rebate written into the amendment	Yours (the brief already chose the amendment)
Ad levy	30% of each company's U.S. digital ad revenue above $5 billion: measurable once companies have to report it, and it raises about what round one's targeted base did ($201 against $197 a person)	A judgment about what can be measured; rising rates are yours; the courts are legal

Switch	What holds up	Settled by
Coverage	S&P 500 first, then all U.S.-listed; private companies later	Law and missing data

What it does, 2025 dollars	2030	2040	2050
Ownership dividend per person	$464	$891	$972
Whole check per person (dividend, ads, carbon)	$1,386	$1,984	$2,291
Share of covered companies the 2020 owners still hold	86%	75%	75%

It costs the 2020 owners a quarter of each covered company, once, and nothing after 2040 on any path tested. Its worst year is a replay of 2008-2009 onto the full fund: the dividend falls 13.5% in one year and about 30% over four, while the companies' own payouts fall 60%. It never sells a share to pay a check, and nobody waits to be paid.

What it gives up is a stake for workers in their own company. Every way of giving them one needs new shares forever, makes leavers wait decades, or eats the citizen check, except one: leavers keep their shares and the shares still count toward the quarter. That one passes too, but it halves the ownership dividend, $445 instead of $891 in 2040. Whether that trade is worth it is your call, along with the lock level, the carbon rebate size, whether ad rates climb with company size, the reserve size inside the close range, the pass lines themselves, and private and foreign companies.

The workbook, Everybody_Owns_a_Piece_Round_Two.xlsx, has every switch live except the per-person cap, which runs in the engine only, and matches the engine to the cent on all 27 combinations checked. The rulebook has the winning setup on a few pages; an outside rebuild from it alone hit all 25 of its check values.

The fixes first

Every fix the reviewers asked for is built in, measured on round one's own settings so the change is easy to see. The bookkeeping still holds; here is what changed.

Fix	What changed	The number now
Who paid	$85 billion of the $2.65 trillion only offset the trusts' own dilution. The $220 billion from companies joining the index is a change in what's measured, shown as a memo	Net transfer 2021-2025: **$2.56 trillion**. Foreign investors $683B, top 0.1% $359B, next 0.9% $435B, next 9% $729B, next 40% $328B, bottom half $29B
Coverage	Shown as covered under the central estimates, with the reserve as the backstop	1.04x to 1.36x. With net buybacks 10% lower, 2025 is 0.98x; with fund costs at 0.2%, 0.99x; with both, 0.93x. The reserve covers every case and no shares get sold
Worker-trust dividends	The $146 billion over five years now has a rule: it goes to current workers or pays people who leave, depending on the exit switch	Joe's slice at Walmart: $58 to $238 a year while he works there, because Walmart pays out only 1.8% of its value
Day-one drop	Labeled a valuation scenario (markets pricing the whole path at once), not bookkeeping, with where the money goes beside it	28% on round one's design, 40% with real turnover, 27% under the winning setup. Winning setup: of the $16.0 trillion, $10.7 trillion goes to the citizen fund (about $31,000 a person, in today's value) and $5.3 trillion to workers through the profit share. Round one's design: $16.6 trillion, $5.7 trillion to each trust and $5.3 trillion through the profit share. 29% of the fall lands on foreign holders
Pay offset	Tested part of the profit share coming back out of base pay, the way it did for France's high-skill workers	If 14% or 50% comes back out, the company's cost falls from 9% of profit to 7.7% or 4.5%, and the fund line rises 1% to 5% by 2040. Joe keeps his; the developer's $45,000 mostly replaces base pay

One correction turned up that nobody flagged. Treasury's $583-a-person rebate was its net revenue, after the 25% offset, so a middle family's carbon cost is 43% of the gross fee per person, not 57%. Round one overstated Joe's family's carbon cost by a third: $1,133 in 2025, not $1,511.

The government's side, in 2025 terms, as if the plan were running:

Line	$ billions a year
Carbon fee: other taxes fall when the fee raises prices (Treasury's 25% offset)	-56
Profit share: corporate tax lost, since it's deductible pay (upper bound, all corporations)	-77
Profit share: income tax workers pay on it (CBO's 18% average marginal rate on labor)	+66
Profit share: payroll tax, only if the law counts it as wages (CBO's 9%)	+33
Ad levy: the same 25% offset	-17
Dividend and gains tax the trusts don't pay (27% of stock sits in taxable accounts)	-6

The carbon line is the choice the brief asked about. Rebate every gross dollar and the Treasury absorbs about $56 billion a year, $80 billion by 2040. Rebate 75% and the carbon line breaks even, with a check a quarter smaller; the other lines still net to about -$33 billion in 2025 without payroll tax. That one is a value choice, covered under Switch 4. Joe's family over forty years has its own section below.

Put together over the five backtest years, the same lines run: carbon offset -$249 billion, the profit share's corporate tax -$352 billion (the reviewers' figure), income tax workers pay on the profit share +$302 billion, payroll tax +$151 billion if the law counts it as wages, the ad levy's offset -$67 billion, and the trusts' untaxed dividends -$16 billion. Net: -$383 billion over 2021-2025 without payroll tax, -$232 billion with it. That's 0.3% or 0.2% of GDP a year.

How the switches were judged

These four pass lines went into this report before a single switch ran. A setup has to clear all four, on the central path and in every stress test, before it gets compared on anything else.

Pass line	A setup passes if	Why the line sits there
How far a check can fall	The ownership dividend falls no more than 15% in any one year after inflation, and the whole check (dividend, ads, and carbon when it's in) no more than 10%	Big enough to ride out an ordinary bad year, small enough that a family wouldn't have to change its plans. A repeal of carbon is a political event, not a market fall, so it's shown separately
Minimum reserve	Zero. The fund can spend its savings down to nothing, but never sells its own shares to pay checks	Every share the fund sells gets replaced with a new one, so selling to pay checks is issuing forever by another name
Most extra issuance	The 2020 owners still hold at least 73% of the companies in year 30 and 70% in year 100, and no stress year needs more than 0.5% of a company in new shares	The quarter alone leaves them 75%. Two points of slack allows small top-ups, not a permanent stream
People who leave get paid	Everyone who quits or retires gets every vested share paid within six years: payments start within a year and run no more than five	That's the schedule employee-ownership law sets for retirees (26 U.S.C. 409(o)). Ordinary plans can make quitters wait five more years; this plan holds them to the retiree schedule

The lines are mine, set before the runs, and each one is a value choice. The workbook lets anyone move them and see what changes.

The setups that pass all four are then compared on the five tests from the brief:

1. Pays the most to the most people: the ownership dividend per person in years 20 and 30, and how many people the ownership money reaches.
2. Costs existing owners the least for it: what the 2020 owners still hold in years 30 and 100, and dollars reaching people per dollar owners give up.

3. Keeps paying in bad years: the worst one-year fall across every stress test.
4. Doesn't need new shares forever: new shares needed each year once the trusts are full.
5. Leaves a worker who quits or retires with what they earned: the share of vested balances paid on time, and anything forfeited.

When one setup beats another on every test, the numbers settle it. When two trade, say one pays workers more and the other pays everyone more, that's a value choice, and it's marked as Chris's call.

Switch 1: what happens to a worker's shares when they leave

Three settings clear every line: the whole quarter in the citizen fund, a pooled worker fund that pays income only, and leavers keeping shares that still count toward the quarter. The first pays the citizen fund twice what the other two do. Everything that pays leavers out of a company trust fails, and for one reason.

People leave covered companies fast. Fitted to BLS's January 2026 tenure count, about 23% of covered workers leave in a year; big-firm separation rates in BLS's job turnover survey ran 18% to 23% in 2021-2025. Most leavers are new hires holding little, but weighted by shares, about 12.2% of a worker trust's value walks out every year. The trust's own dividends and buybacks bring in 2.7%. The other 9.5% has to come from somewhere every year: new shares, a waiting line, the companies' cash, or nowhere, because the leaver gets nothing to cash out.

Every setting below ran with the cash payout rule from Switch 2. Under the essay's 3% rule nothing passes, so that rule can't separate them.

Setting: what a leaver gets	Result	Ownership dividend, 2040	New shares a year once full	Joe, laid off after 10 years at Walmart
Whole quarter to the citizen fund; workers keep the profit share (F)	Passes	$891	0	No trust stake. His $5,568 of profit share over 2021-2025 is already his

Setting: what a leaver gets	Result	Ownership dividend, 2040	New shares a year once full	Joe, laid off after 10 years at Walmart
One pooled worker fund that pays current workers its income; leavers keep nothing (G)	Passes	$445	0	$6,569 of income while he works there, nothing after
Leavers keep their shares and the quarter counts them for good (A_forever)	Passes	$445	0	Keeps $13,559 of Walmart stock, paying him about $238 a year
Leavers keep their shares, counted only while held (A_hold)	Fails: new shares forever	$457	0.82%	Same as above
The trust buys leavers out from its own cash only (B)	Fails: leavers wait 30 years or more	$445	0	$14,473, paid in 2055
The company pays leavers cash (C)	Fails: the check collapses and the fund sells shares	$175	0	$18,941 from Walmart in 2026
The trust pays leavers and gets topped up with new shares (D)	Fails: new shares forever	$457	0.85%	$14,327 in 2026, and no trust cash while he works there
D with a three-year vesting cliff	Fails: new shares forever	$457	0.76%	$17,376 in 2026
D with a $100,000 cap per person, overflow to the citizen fund	Fails: new shares forever	$678	0.24%	$14,327 in 2026

Setting: what a leaver gets	Result	Ownership dividend, 2040	New shares a year once full	Joe, laid off after 10 years at Walmart
D with both the cliff and the cap	Fails: new shares forever	$736	0.18%	$17,376 in 2026
D with a fifth of the new shares to workers instead of half	Fails: new shares forever	$731	0.29%	$5,731 in 2026
One portable account across every covered company; 10% of leavers land at another one (E)	Fails: new shares forever	$457	0.78%	$68,969 in 2026
E with 30% landing at another covered company	Fails: new shares forever	$457	0.67%	$69,152 in 2026

Ownership dividend: per person, 2025 dollars. New shares: the average a year from 2050 to 2120, as a share of each company. The pass line keeps the 2020 owners at 73% or more in year 30 and 70% or more in year 100, which a permanent stream of this size breaks. Joe: round one's warehouse worker, laid off at the end of 2025; 2025 dollars.

What the variations show:

- Vesting and caps shrink the leak; they don't stop it. A three-year cliff forfeits the shares of the 68% of leavers who go before three years, and those shares go back to the people still there. A $100,000 cap moves most of the worker half to the citizen fund: with nobody leaving, the worker trusts would hold 3.9% of the companies by 2040 instead of 12.5%. What's left still leaks.
- Company trusts pay by company value, not by work. At the quarter, round one's numbers give a Walmart worker about $49,000 of stock and an Nvidia worker about $12.4 million. The portable account evens that out (Joe's account would be $69,000 instead of $14,300), but it still needs new shares forever.
- Under D the trust's dividends never reach the people still working. All of it goes to leavers, and it still isn't enough. That answers where the $146 billion went.

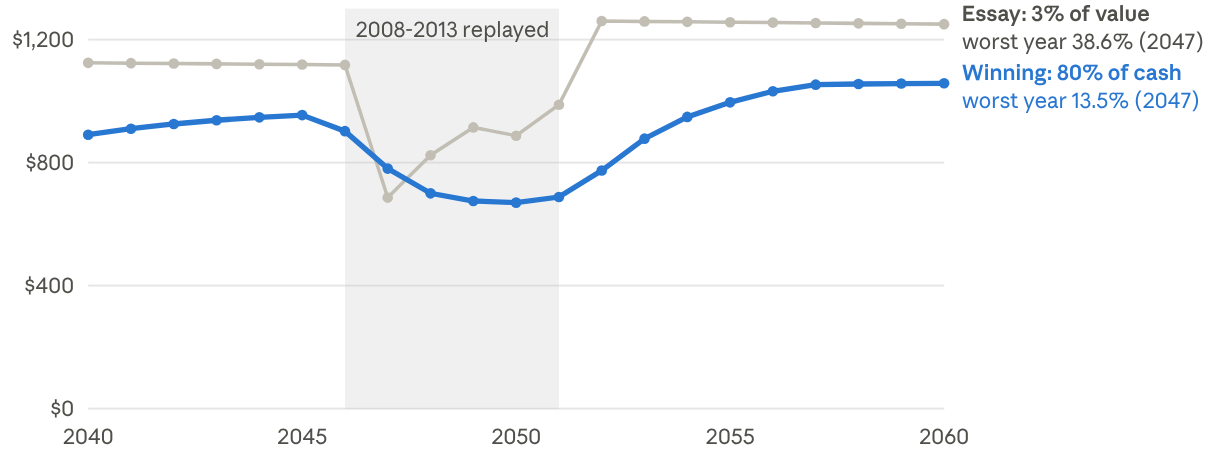
The fifth test, leaving with what you earned, is where the numbers stop. Under the winning setting nothing is held back from a worker: the profit share is paid every year, $850 to $1,400 for Joe, half to spend and half into his own locked account, and all of it is his. What a worker doesn't get is a piece of the company. Only A_forever gives every worker a piece they keep, and it passes every line, at half the citizen dividend. That one's yours.

Switch 2: the payout rule

One rule of the ten passes everything: pay out 80% of the fund's share of the cash companies paid owners, averaged over the last five years, and never more than the cash in hand plus the reserve. The essay's 3% of value fails because companies hand owners only 2.7% of their value in cash, so a fund paying 3% has to sell shares every year from 2028 on.

In a 2008-style crash the essay's rule cuts the check 39% in one year; the cash rule, 13.5%

Ownership dividend per person, 2025 dollars. Fund full since 2040; 2008-2013 markets replayed from 2046. Pass line: no year falls over 15%. The essay's rule pays more in normal years by selling shares.



Engine runs (engine.py, scenarios.py): S&P 500, 2008-2013 real price and payout changes applied in 2046

The cash rule pays less early ($464 in 2030 against $642 for the essay's rule) because it pays 80% of a lagging average and builds a reserve first. By 2040 it holds about $690 billion, two years of checks, and that reserve is what holds the 2008 replay to a 13.5% worst year. Above two years, a fifth of the extra is paid out each year as a bonus.

Rule	Result	2030	2040	2050	Worst one-year fall	What breaks
3% of Jan 1 value (the essay)	Fails	$642	$1,125	$1,113	41%	Sells shares 93 years out of 95; the 2020 owners drop to 67.7% by 2120
3% of the five-year average value	Fails	$645	$1,125	$1,113	33%	Sells shares every year from 2031
Lesser of 3% of value and the cash that came in	Fails	$569	$993	$983	44%	No cushion: the check falls as far as the cash
2.5% of Jan 1 value	Fails	$538	$937	$928	39%	Sells shares when the cash dries up
2.5% of the five-year average value	Fails	$538	$937	$928	35%	Same
Lesser of 3% of average value and cash in	Fails	$569	$993	$983	44%	No cushion
3% of average value, capped at cash in plus reserve	Fails	$645	$993	$983	44%	The rule asks for more than the cash, so the reserve never builds
2.5% of average value, capped at cash in plus reserve	Fails	$538	$937	$928	35%	The reserve bonus is paid all at once, then stops all at once
The same, bonus paid a fifth a year	Fails	$538	$937	$928	25%	Based on value, so it can't see cash drying up while values hold; the reserve runs out and the check drops to the cash
80% of the five-year average cash, capped at cash in plus reserve	Passes	$464	$891	$972	13.5%	

Ownership dividend per person, 2025 dollars, all with the whole quarter in the citizen fund. Worst one-year fall across all ten market paths.

The neighbors of the winning rule are a close call:

Variant	Result	2030	2040	Paid out 2026-2060	Worst one-year fall	2008 slide, peak to trough
75% of average cash	Passes	$435	$896	$9.97 trillion	13.4%	29%

Variant	Result	2030	2040	Paid out 2026-2060	Worst one-year fall	2008 slide, peak to trough
80%, two-year reserve (winning)	Passes	$464	$891	$10.00 trillion	13.5%	30%
85%	Passes	$493	$860	$10.02 trillion	14.3%	30%
90%	Fails	$522	$910	$10.05 trillion	15.3%	
80%, one-year reserve	Passes	$494	$934	$10.28 trillion	14.6%	34%
80%, three-year reserve	Passes	$464	$809	$9.71 trillion	12.1%	24%
80%, a tenth of the excess paid a year	Passes	$464	$857	$9.72 trillion	11.3%	30%
80%, the whole bonus at once	Fails	$464	$990	$10.25 trillion	19.4%	

Paid out: all ownership dividends, 2025 dollars. The rate barely matters past 2040; the reserve size is the real lever. A one-year reserve pays 3% more over 35 years and lets a 2008-style slide reach 34%; a three-year reserve pays 3% less and holds it to 24%. All of them pass the line set in advance, so the numbers can't pick one. Two years sits in the middle. How hard a crash should land on a family's check is yours.

Switch 3: the lock

The numbers lean hard on one piece: the carbon check should be fully spendable. It exists to cover higher prices, and those come due now. With half of it locked, the essay's rule, a middle family clears only $44 a person a year after the higher prices with the full rebate, and falls $38 a person short every year with a 75% rebate. Unlocked, it clears $371 or $208.

Past that, the lock is a trade between money now and money built, and the numbers only show the trade:

Lock rule	Spendable 2025	Spendable 2040	Building 2040	Building 2060	House, 10% down	Each kid takes at 18
Carbon spendable, half of the rest locked (split)	$3,026	$2,105	$21,213	$52,378	2043	$4,877 and $6,813
Half of every check (the essay)	$1,717	$1,208	$36,487	$106,071	2035	$11,293 and $15,119
A quarter of every check	$3,142	$2,200	$20,318	$51,529	2044	$5,646 and $7,560

Lock rule	Spendable 2025	Spendable 2040	Building 2040	Building 2060	House, 10% down	Each kid takes at 18
No lock	$4,567	$3,192	$0	$0	Not by 2060	$0

Joe's family under the winning setup, 2025 dollars. Spendable is what the household can spend that year after the carbon prices. Building is the parents' locked account plus the home equity from the down payment; the kids' accounts leave with them. The ad check and the profit share stay under the lock with the ownership dividend.

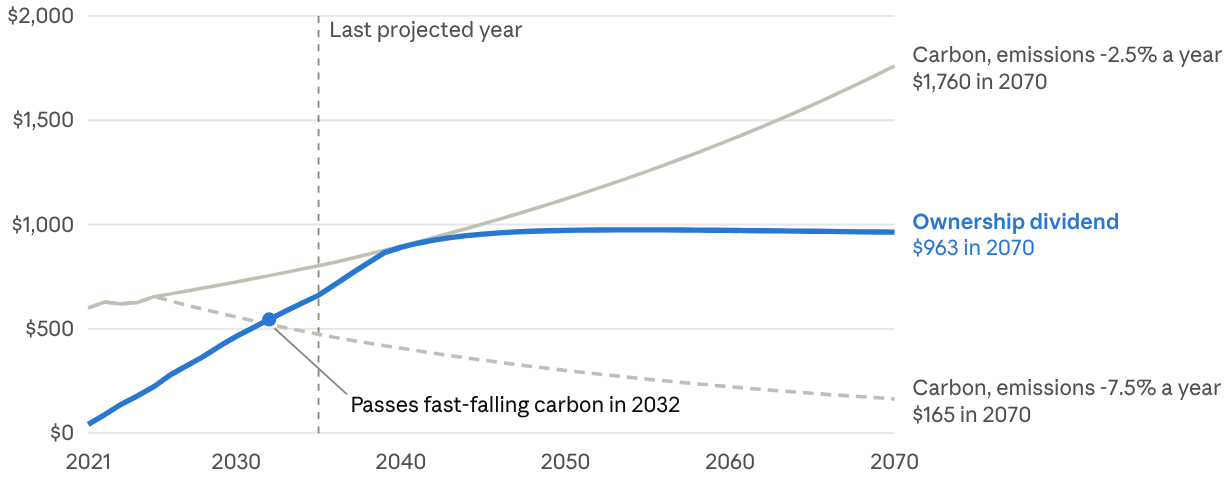
The split rule and a quarter lock land within a few percent of each other on every column but the kids' accounts, which the split rule leaves 11% to 16% smaller. The difference is timing: under the split, the carbon money arrives as cash when the prices go up. Half of every check builds twice as much and buys the house eight years sooner, at about $1,300 a year less to spend now. How much a family should be made to save is yours.

Switch 4: carbon, with and without it

The carbon check stays the biggest piece of the money through 2040 and keeps growing after, as long as the fee climbs faster than emissions fall. Under the winning setup with the full rebate, the ownership dividend draws level with it around 2040 ($891 against $897) but doesn't pass it. With a one-year reserve it runs ahead from 2038 to 2043 ($934 against $897 in 2040), and with a 75% rebate it passes in 2033 and stays ahead until 2056. If emissions fall 7.5% a year instead of 2.5%, it passes in 2032 for good. The essay's 3% rule gets past carbon by year twenty only with the whole quarter in the citizen fund, and only by selling shares; the essay as written ($571 in 2040) never does.

The dividend catches carbon around 2040 but passes it only if emissions fall faster

Per person a year, 2025 dollars, winning setup. Fee $43 a ton in 2021, rising 5% a year above inflation.
Full rebate shown. No source projects the fee path past 2035; later carbon checks extend the rule.



Engine runs (engine.py): carbon path from Baker-Shultz fee and RFF emissions ratio (S14, S15), EIA emissions (S13), CBO population (S38)

The floor without carbon (no fee, no rebate, no higher prices) is the ownership dividend plus the ad check: $420 a person in 2025, $661 in 2030, $1,087 in 2040 and $1,169 in 2050. With carbon the whole check is $1,074, $1,386, $1,984 and $2,291.

Writing the rebate into the amendment, as the brief says, protects it from a repeal like Canada's: it ended its consumer carbon price in April 2025, a year after its finance department said 8 in 10 families got back more than they paid. The amendment still has to say how much of the fee goes back:

Rebate	Carbon check per person, 2025	Carbon line in the federal budget, 2025	Middle family's carbon money after higher prices, per person
All of the fee, tax-free	$654	-$56 billion a year, -$80 billion by 2040	+$371
75% of the fee, tax-free	$491	About even	+$208
All of the fee, taxable and open to everyone	$654 before tax	No offset under JCT's scoring	+$371 less the family's income tax on it

The offset is real money. When the fee raises prices, taxable income and payroll fall, and Treasury's own study charges 25% of the gross fee for it; round one never did. Which route to take is a value choice: a bigger tax-free check the rest of the budget pays for, a smaller one that pays for itself, or the full check with income tax taken out.

Switch 5: the ad levy base

The levy on "targeted" ads rests on a number nobody measures: what share of ads count as targeted. Round one used 80% because that's what the essay's $200 needs. Rising rates on each company's U.S. ad revenue, the design the brief named (Paul Romer's), don't need that number, and they raise more.

Base	Levy, 2025 ($ billions)	Check per person, 2025
30% of targeted ads, 80% counted (round one)	67	$197
30% of targeted ads, search left out	51	$150
Romer's default rates: 0% under $5 billion, rising to 72.5% above $60 billion	86 to 130	$252 to $380
Romer's alternative, topping out at 50%	80 to 111	$235 to $326
Flat 30% above $5 billion	59 to 79	$172 to $231
0%, 20%, 35% steps	59 to 81	$172 to $237

No company reports U.S. ad revenue by itself, so Google, Meta and Microsoft are each a low-high range built from their filings and industry estimates, and Amazon and TikTok are single estimates (the Ad levy tab shows each one). Only those five clear $5 billion; every other seller pays nothing under the rising-rate designs. All figures assume 95% collected.

Three things here aren't arithmetic:

- Romer says the point of his rates is to change how platforms make money, not to raise revenue. If it works, the levy shrinks. The smaller-base test (25% less from 2026) costs $49 a person a year, and the whole check still never falls in a year because the ownership dividend is still climbing then.

- Part of the levy comes back out of shoppers. In Europe, Amazon passed about half of the digital services taxes on to its sellers, and a tax on Facebook raised its ad prices 25% to 38%.
- The courts. Maryland's ad tax was struck down by its Tax Court on August 14, 2026 and is on appeal; Illinois's and Utah's start in 2027 and are already being sued. A federal levy stands on different ground, but whether it stands is a legal question.

One piece is a judgment rather than a number. A base built on "targeted" is hard to hold for thirty years, because every year would be a fight over what counts. A flat 30% on each company's U.S. digital ad revenue above $5 billion raises about what round one's targeted base did ($201 a person in the middle of the range, against $197) on a base that's measurable once companies have to report it, the way Maryland's tax made them. So the winning setup uses that base. The totals in this report and the pass-line runs still carry round one's $197 line; the gap is about $5 a person. Whether the rate should climb with company size, as Romer wants, is yours.

Switch 6: who's covered and the foreign formula

Covering every U.S.-listed company instead of just the S&P 500 raises the ownership dividend about 18%. The foreign formula, issuing 1.5% times each company's U.S. share of revenue, cuts it by about the share it removes: roughly 28% at S&P's estimate that 71% of S&P 500 revenue is American.

Coverage	Ownership dividend 2030	2040	2050	2020 owners keep
S&P 500 (the main run)	$464	$891	$972	75%
All U.S.-listed companies	$548	$1,046	$1,146	75%
Foreign formula, 71% U.S. revenue	$336	$638	$691	82%
Foreign formula, 57% U.S. revenue	$273	$515	$556	86%

The formula is read here as claiming a quarter of each company's U.S. business, so both the 1.5% and the quarter scale with the U.S. share. Estimates of that share for the S&P 500 run from 57% to 72% depending on who measures it and how (S&P, Goldman Sachs, FactSet), and they don't reconcile.

Private companies can't be run from public data. The Fed counts $15.6 trillion of closely held company stock of all sizes, $12.1 trillion of it in S corporations. Census counts 21,041 firms with 500 or more employees, but that count mixes public, private, foreign-owned and nonprofit employers, and nobody publishes what the private ones above the size line are worth. The cash option, where a private company pays the fund each year what its new shares would have been worth, needs a yearly valuation of every covered private company. That's an administrative and legal design question, and putting a dollar figure on it here would mean making one up, so there isn't one. Foreign companies' U.S. arms employed 8.6 million people and sold $6.5 trillion in 2024; covering them runs into the same valuation problem plus the treaties.

The winning setup covers the S&P 500 first and all U.S.-listed companies next. Whether to apply the foreign formula is a legal judgment (it narrows the claim on foreign owners and foreign-earned value, which matters if the plan is challenged under a tax treaty) and a value choice about how much of the check to give up for that protection.

Stress tests

The winning setup passed every test the brief named, plus four harsher ones added after the first runs. Its worst case is 2008-2009 replayed onto the full fund: the dividend falls 13.5% in the worst year and about 30% over four years, while the cash companies paid their owners fell 60%. The fund never sells a share. Once it's full, its reserve never drops below $529 billion. The closest call comes earlier: the cash squeeze hits while the fund is still filling and runs the reserve down to $10 billion in 2030, about a month of checks. The line holds there, barely.

Path	Worst one-year fall	Peak to trough	Ownership dividend, 2050	Sells shares
Central: no real growth, 2025 cash yield	0.1%	1%	$972	No
2000-2002 replayed onto the full fund in 2046	1.4%	3%	$984	No
2008-2009 replayed	13.5%	30%	$670	No
Falling profits and buybacks, heavy worker withdrawals	11.6%	23%	$752	No

Path	Worst one-year fall	Peak to trough	Ownership dividend, 2050	Sells shares
AI boom: values up, jobs down, cash to data centers	0.1%	1%	$3,361	No
Prices fall back to the long-run average valuation	0.1%	1%	$982	No
2000-2002, half of buybacks offset by stock pay	0.9%	1%	$995	No
2008-2009, half of buybacks offset by stock pay	11.5%	25%	$715	No
Cash squeeze without a boom	4.3%	7%	$967	No
Average valuation first, then 2008-2009	13.4%	30%	$679	No

Per person, 2025 dollars. The 0.1% falls on the calm paths are population growth: the same money spread over more people.

How the paths were built. In the replays the fund is full by 2040, and in 2046 it meets the actual year-by-year changes in real stock prices and in the cash companies paid owners during 2000-2005 or 2008-2013. Buybacks before 2021 are only published gross, so net buybacks are taken as 90% of gross (the 2021-2025 measured ratio) or, in the harsher versions, 50% (the late-1990s finding that stock pay offset about half of buybacks). The falling-profits path is illustrative: values down 35%, payouts down 30% two years running, layoffs adding 10 points to separations, covered jobs down 10%. The AI path takes BlackRock's AI-boom case for values (13.2% a year after inflation to 2035), cuts cash payouts to 1.5% of value while companies build, and drops covered jobs 3% a year to 2035; the payout cut and the job loss are illustrative.

The carbon and ad tests:

- Fast decarbonization, emissions down 7.5% a year (the pace of the 2035 U.S. climate pledge): the carbon check shrinks every year from 2026, but slowly enough that the whole check never falls more than 0.6% in a year. It's $1,494 a person in 2040 instead of $1,984.
- A smaller ad base, 25% less from 2026: the ad check drops $49 a person, the whole check doesn't fall in any year, and 2040's is $1,935.

Heavy worker withdrawals don't touch the winning setup, which holds no worker trust. They do bite the designs that have one: a portable account with 30% of leavers staying covered needs 2.6% of each company in new shares in its worst year, five times the line.

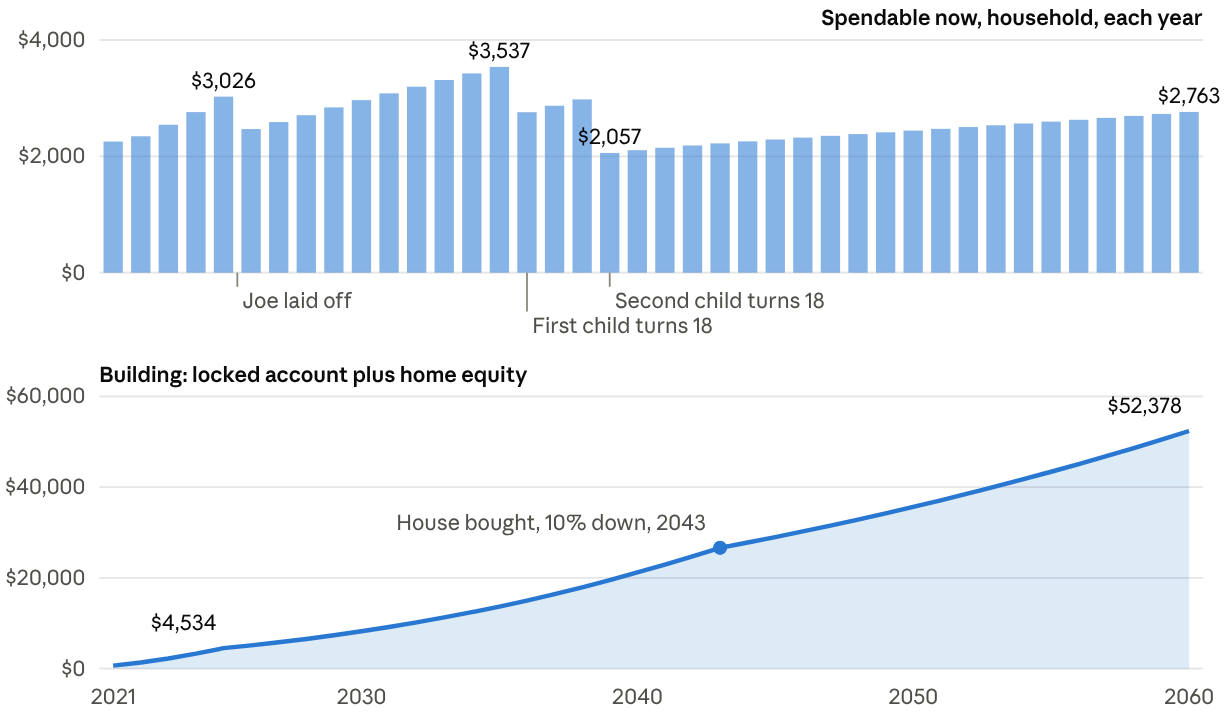
What these tests can't cover: a slump longer than anything in the record, a permanent drop in the share of profits companies pay out, or companies holding back payouts to starve the fund. The cash rule passes all of those straight through to the check, because it pays out what comes in.

Joe's family to forty years

Under the winning setup Joe's family has $2,000 to $3,500 a year from the plan to spend in every one of the forty years, and builds $52,000 by 2060, enough for a down payment on a starter home in 2043.

Joe's family spends $2,000 to $3,500 a year from the plan and builds $52,000 by 2060

Winning setup, carbon checks spendable and half of the rest locked; 2025 dollars. An illustration, not a forecast. Spendable is after the higher prices carbon brings. Building is the parents' locked account plus home equity.



Engine runs (family.py): winning setup, split lock; Walmart data (S19, S22, R2-03); Redfin and NAR (R2-11, R2-12)

The story is round one's, carried out to 2060. Joe is 27 in 2021 and has worked at Walmart since 2016 for $45,000. His wife works part time at a company too small to be covered. Their kids were born in 2018 and 2021. Walmart automates and lays Joe off at the end of

2025; from 2027 he works for a smaller, uncovered employer at the same pay. When the parents' locked account reaches 10% of a $260,508 starter home (Redfin's August 2025 median; 10% is the median first-time buyer's down payment), they buy. Each kid takes his or her checks and locked account at 18. Locked money earns 3% a year after inflation, the essay's figure.

What moves the line:

- The profit share, $850 to $1,400 a year in 2025 dollars while Joe is at Walmart (half to spend, half into the locked account), stops with the layoff. Spendable money drops from $3,026 in 2025 to $2,468 in 2026, then climbs back with the ownership dividend.
- When each kid turns 18 the household loses a set of checks: $3,537 to spend in 2035, $2,057 by 2039. The kids leave with $4,877 and $6,813 in their own locked accounts.
- After 2040 the dividend holds steady and the carbon check keeps growing, so spendable money creeps back up to $2,763 by 2060.

Without carbon the family's spendable money is $1,542 in 2025, $1,087 in 2040 and $1,169 in 2060; what it builds doesn't change, because the carbon check was all spendable. Under the essay as written (a worker trust topped up with new shares, the 3% rule, half of everything locked) the family builds faster, $40,642 by 2040, mostly because Joe's $14,327 trust stake is paid out at the layoff and half of the carbon check is locked. That design needs new shares forever and pays everyone else less. The Joe 40 years tab in the workbook has every year, live, under whatever setup is on the Switches tab.

The winning setup, and your calls

Here is the setup, piece by piece, with what settles each one. With these settings the plan passes every line on every path tested:

1. No worker trust. The whole quarter goes to the citizen fund; workers keep the profit share, 10% of pre-tax profit capped at three months' pay, paid to them every year. The numbers rule out every design that pays leavers from a trust, makes them wait or has companies pay them; of the three that pass, this one pays everyone twice as much. Whether workers should also own a piece is your call 1 below.
2. The cash payout rule: 80% of the fund's share of the last five years' company payouts, never more than cash in plus the reserve, the reserve capped at two years of checks with a fifth of any excess paid out each year. Settled by the numbers, inside the close range in your call 5.

3. The carbon check fully spendable. The numbers lean hard on it: half locked, a middle family nets $44 a person, or falls short with a smaller rebate.
4. A measurable ad base: each company's U.S. digital ad revenue above $5 billion. A judgment about what can be measured, not a number.
5. The S&P 500 first, then every U.S.-listed company. Law and missing data decide the rest.

The rulebook writes these out so someone who didn't build the model could rebuild it.

Your calls. Each one is a value choice; the workbook shows what every setting does.

1. Whether workers get a stake of their own. Leavers keeping shares that still count toward the quarter passes every line too, at half the ownership dividend: $445 against $891 a person in 2040.
2. How much of the rest of the check to lock: split, a quarter, half, or none.
3. The carbon rebate: all of the fee tax-free (the Treasury covers about $56 billion a year), 75% (it pays for itself), or all of it taxable.
4. Whether the ad rate stays flat at 30% or climbs with company size the way Romer designed it.
5. The reserve: one, two or three years of checks, which trades about 3% of total payouts either way against how far a 2008-style slide goes (34%, 30% or 24%).
6. The pass lines themselves.
7. The foreign formula, and when and how private companies come in. Both are legal questions first.

What's left

Call it: this is as good as it gets with what's known. The winning setup is the best of everything tested against lines set in advance, it holds on every path, every number in it traces to a source or a stated rule, and what's left can't be settled by running more numbers.

What doesn't hold, said plainly:

- The essay's 3% payout. Companies hand owners 2.7% of their value in cash, so a fund paying 3% sells shares nearly every year.
- A worker trust that pays leavers out. It needs new shares forever.

- The fund taking over from carbon by year twenty. With the full rebate and the winning rule, the dividend only draws level around 2040; it gets ahead only with a smaller reserve, a smaller rebate, or emissions falling faster than the fee path assumes.

What's left open:

- Behavior. Whether companies hold back payouts once a quarter of their cash goes to the fund. The cash rule passes any squeeze straight to the check (the cash-squeeze test takes 27% off the 2030 check), and nobody can measure that intent in advance. The same goes for how much of the profit share comes back out of pay (France says almost none, except at the top), how ad platforms respond, and how markets would price the plan on day one.
- Law. A tax exemption for a federal citizen trust; challenges to the share issue under takings law and tax treaties; whether the profit share counts as wages for payroll tax; the courts on ad taxes.
- Data nobody publishes. The value of private companies above the size line, each firm's U.S. ad revenue, what share of ads is targeted, net buybacks before 2021, and S&P's fourth-quarter 2025 buybacks (set equal to the third quarter here).
- Carbon past 2035. No source projects the fee path that far; later carbon checks extend the plan's own rule.

The headline numbers, from the dividend path and the budget lines to Joe's house, are labeled measured, calculated, illustrative, behavior, legal or value choice on the workbook's Number labels tab. The Government tab adds up the five backtest years line by line.

Two checks were run on the finished work. Someone working from the rulebook and the raw data alone rebuilt all 25 of its check values, and found six places where the wording allowed more than one reading; the rulebook now says which. A line-by-line fact-check of this report against the engine's output turned up about twenty slips and overstatements, from a path count to claims that called a judgment settled; all are corrected here.

Sources

Round two added 41 sources (R2-01 to R2-41) to round one's 55 (S01 to S55). The table lists every round-two source and the round-one sources this report leans on. Links are on the Sources tab of the workbook.

ID	What it supports	Source
R2-01	S&P 500 market value, dividends and gross buybacks 2012-2024 (and 2016-2020 used for five-year averages)	S&P Dow Jones Indices buyback press releases, annual tables
R2-02	CPI-U annual averages 1995-2016; December-to-December inflation 1997-2013	BLS via FRED (CPIAUCNS); Damodaran histretSP.xls
R2-03	Cash paid to owners (dividends + share repurchases) by Walmart, Amazon, Target, Costco, Kroger, Home Depot, Apple, Nvidia	Company 10-K cash flow statements on SEC EDGAR, cross-matched to XBRL companyfacts
R2-04	The 25 percent offset: an excise tax shrinks the income and payroll tax base; Treasury rebate of $583 per person came from net revenue	U.S. Treasury OTA Working Paper 115 (Horowitz et al., Jan 2017); JCT JCX-59-11 (Dec 23, 2011); CBO (Jan 13, 2009); JCT JCX-10-25 (Feb 2025)
R2-05	S&P 500 companies: market value and latest annual headcount, firm by firm (29,672,422 employees worldwide)	Stock Analysis S&P 500 list (Oct 16, 2026 close) and company employee pages (citing filings); six counts replaced from 10-Ks
R2-06	Share of U.S. corporate equity held in taxable accounts: 27% (2022)	Rosenthal and Mucciolo, "Who's Left to Tax?", Tax Notes Federal, Apr 1, 2024 (Tax Policy Center)
R2-07	Marginal tax rate on labor income: 27% in 2018 (18 income, 9 payroll); payroll 6.2% + 1.45% each side, 2025 wage base $176,100	CBO, "Marginal Federal Tax Rates on Labor Income: 1962 to 2028" (Jan 24, 2019); SSA
R2-08	S&P 500 as-reported earnings, 12 months to Sep 2025: $1,994.76 billion	S&P Dow Jones Indices, Q3 2025 buyback release (Dec 18, 2025)
R2-09	S&P 500 1997-2011: year-end market value, dividends, gross buybacks, earnings, price and total return	S&P DJI release tables (2007-2011); S&P-sourced charts digitized from Yardeni Research and ECB Financial Stability Review Dec 2005 (1998-2006); Slickcharts returns

ID	What it supports	Source
R2-10	Damodaran implied-premium data: S&P 500 year-end levels (1257.60, 1426.19, 1848.36 for 2011-2013), dividends plus buybacks per index unit	Aswath Damodaran, histimpl.xls (saved Jan 2026)
R2-11	Starter-home median sale price $260,508 (Aug 2025)	Redfin, starter home report
R2-12	First-time buyers: 21% of buyers, median age 40, median down payment 10%	NAR, 2025 Profile of Home Buyers and Sellers (Nov 4, 2025)
R2-13	Tenure with current employer, January 2026: distribution (Table 3) and medians (Table 1); median 4.1 years	BLS, Employee Tenure in 2026 (USDL-26-1532, Sept 24, 2026)
R2-14	Separations at establishments with 5,000+ employees, 2021-2025 (yearly rates computed from monthly data): 22.7% to 18.2%	BLS JOLTS database (file dated Sept 29, 2026)
R2-15	Where workers leaving 500+ employee firms go (2024): 34.0% to another 500+ firm, 18.9% to a smaller private firm, 2.5% public sector, 41.0% persistent nonemployment	Census LEHD Job-to-Job Flows, release R2026Q3
R2-16	Vesting and payout law for employee ownership plans	26 U.S.C. 411(a)(2)(B); 26 U.S.C. 409(o), 409(h)
R2-17	ESOPs: 6,609 plans, 15.1 million participants, $2.06 trillion; companies buy back a median 5% of shares a year	NCEO, Employee Ownership by the Numbers (Jan 2026); 2023 repurchase obligation survey

ID	What it supports	Source
R2-18	France's mandatory profit sharing: total pay rises with no base-pay cut on average; high-skill workers' base pay falls by about what they receive (about 100% offset); low and intermediate skill about 0%	Nimier-David, Sraer and Thesmar, NBER Working Paper 31804 (revised Aug 2025)
R2-19	Mandatory profit sharing shows zero substitution for wages; voluntary schemes near-complete	Conseil d'analyse economique, Note 79 (July 2023)
R2-20	U.S. evidence: trade-offs between base pay and shared compensation are minimal	Kruse, Freeman and Blasi, Shared Capitalism at Work (NBER, 2010), ch. 8
R2-21	Romer's digital ad tax: rising marginal rates on a firm's U.S. digital ad revenue, 0% under $5 billion up to 72.5% above $60 billion; alternative schedule tops out at 50%; his 2021 firm inputs (Microsoft $6.65B U.S.)	Paul Romer, Taxing Digital Advertising (May 17, 2021) and FAQ (May 6, 2019)
R2-22	U.S. internet ad revenue 2025 $294.6 billion; search 38.8%; top 10 companies about 84%	IAB/PwC Internet Advertising Revenue Report, FY2025 (Apr 16, 2026)
R2-23	Company ad revenue 2025: Meta U.S.-and-Canada ads by user location $85.3B, U.S.-billed revenue $74.78B; Alphabet U.S. revenue 48% of total, Google advertising $294.7B worldwide; Amazon advertising $68.6B worldwide, U.S. about $48B (2026 forecast $56.71B less 17.9% growth); TikTok U.S. about $12B; Microsoft search advertising $15.2B worldwide (FY2026); Google, Meta and Amazon about 72% of U.S. digital ad spending	Meta, Alphabet, Amazon, Microsoft 10-Ks (FY2025, FY2026); EMARKETER (2026); WARC via Marketing Dive (2025); EMARKETER via PPC Land (Aug 29, 2025)

ID	What it supports	Source
R2-24	State ad taxes, Oct 2026: Maryland's struck by its Tax Court (Aug 14, 2026; on appeal); Washington taxes ad services since Oct 1, 2025; Illinois 10% tax on targeted ads from Jan 1, 2027 (sued); Utah 4.70% from 2027 (sued)	Maryland Tax Court, Apple v. Comptroller (No. 23-DA-OO-0456); 4th Cir. No. 24-1727; WA DOR; Illinois P.A. 104-0468; Utah S.B. 287
R2-25	Who bears digital ad taxes: Amazon passed about half of European digital services taxes into seller fees; a tax on Facebook raised its ad prices 25%-38%	Langenmayr and Muddasani (2025); Lassmann et al., European Economic Review 172 (2025)
R2-26	Energy CO2: 2005 6,007.5 MMT; 2025 4,904.4; AEO2026 counterfactual baseline 4,431 (2030), 3,844 (2035), 3,656 (2040), 3,791 (2050)	EIA (July 2026 emissions report; Annual Energy Outlook 2026, Apr 8, 2026)
R2-27	U.S. 2035 NDC: 61%-66% below 2005 (submitted Dec 2024; the U.S. left Paris Jan 27, 2026)	UNFCCC
R2-28	RFF projections with the Baker-Shultz fee: 4.1 (2021), 3.7 (2025), 3.3 (2030), 3.0 (2035) billion tons; without: 5.1, 5.0, 4.9, 4.8	Resources for the Future, Issue Brief 18-07 (rev. Mar 2019), Table 1a
R2-29	Closely held corporate equity $15.637 trillion (S corporations $12.070 trillion), 2026:Q2	Federal Reserve, Financial Accounts Z.1 (Sept 11, 2026)
R2-30	Firms with 500+ employees: 21,041 firms, 73.5 million jobs (54.1%), 2022	Census, Statistics of U.S. Businesses 2022 (Apr 10, 2025)
R2-31	U.S. affiliates of foreign companies, 2024: 8.57 million jobs (6.1% of private jobs), sales $6.518 trillion, value added $1.518 trillion	BEA (July 10, 2026)

ID	What it supports	Source
R2-32	America's Top Private Companies 2025: 200 firms, $2.02 trillion revenue, 3.88 million employees	Forbes (Dec 17, 2025)
R2-33	About half of large S&P 500 firms' gross repurchases in 1994-98 were offset by employee option exercises	Liang and Sharpe, Federal Reserve FEDS 1999-59
R2-34	AI spending and payouts, 2025-2026: Alphabet, Meta zero buybacks in H1 2026; nonfinancial companies net issuers of stock in Q1-Q2 2026; Q1 2026 buybacks +1% vs capex +38%	Company filings; Fed Z.1 via FRED NCBCEBQ027S; Goldman Sachs via Yahoo Finance and Fortune (2026)
R2-35	Tax status: Alaska's dividend is federally taxable; state entities excluded under 26 U.S.C. 115; the federal Railroad Retirement Investment Trust is exempt by name in 501(c)(28)	Alaska PFD Division; 26 U.S.C. 115, 501(c)(28)
R2-36	Shiller CAPE: Dec 1999 44.20, Dec 2000 37.27, Dec 2007 25.96, Dec 2008 15.38, Sep 2026 40.58	Multpl (Shiller data)
R2-37	Canada ended its consumer carbon price Apr 1, 2025; Finance Canada said 8 in 10 families got back more than they paid (Feb 14, 2024)	Finance Canada
R2-38	Mexico's 2021 reform capped profit sharing at three months' pay or the three-year average, whichever is higher	Basham, Ringe y Correa (Apr 4, 2024); IZA DP 18566 (Apr 2026)
R2-39	FHA minimum down payment 3.5% (credit score 580+); 30-year mortgage averaged 6.60% in 2025	HUD; Freddie Mac PMMS (weekly, averaged)

ID	What it supports	Source
R2-40	Federal brackets 2025 (married filing jointly) and 2025 standard deduction $31,500	IRS IR-2024-273; IR-2025-103
R2-41	The round-two engine: the plan as plain code, tenure-based worker model, stress paths	engine.py, scenarios.py, score.py, family.py, hazard.py, analysis.py (in the SOMA folder, engine subfolder)
S01	U.S. resident population, July 1, 2020-2025	U.S. Census Bureau, Vintage 2025 national and state population estimates (NST-EST2025-ALLDATA)
S02	Consumer prices (CPI-U), annual averages	U.S. Bureau of Labor Statistics, CPI-U all items, not seasonally adjusted, via FRED series CPIAUCNS
S03	S&P 500 market value 2020-2024, dividends and buybacks 2020-2024, buyback-tax estimates, buybacks and buyback tax for Q1-Q3 2025	S&P Dow Jones Indices press release, "S&P 500 Q3 2025 Buybacks Post Modest 6.2% Gain to $249.0 Billion" (Dec 18, 2025), data tables
S04	S&P 500 market value at Dec 31, 2025 and 2025 dividends	S&P Dow Jones Indices, Market Attributes: U.S. Equities, December 2025
S05	2024 buybacks ($942.5B), 2024 dividends ($629.6B), 2023-2024 buyback tax	S&P Dow Jones Indices press release, Q4 2024 buybacks (Mar 19, 2025)
S06	2023 buybacks ($795.2B), 2023 dividends, 2023 buyback tax ($7.47B)	S&P Dow Jones Indices press release, Q4 2023 buybacks (Mar 18, 2024)
S07	2022 buybacks ($922.7B), 2022 dividends ($564.6B), pro forma 2022 buyback tax ($8.47B)	S&P Dow Jones Indices press release, Q4 2022 buybacks (Mar 21, 2023)
S08	2021 buybacks ($881.7B), 2021 dividends ($511.2B), pro forma 2021 buyback tax ($7.93B)	S&P Dow Jones Indices press release, "S&P 500 Buybacks Set Quarterly and Annual Record" (Mar 15, 2022)

ID	What it supports	Source
S09	S&P 500 closing levels and calendar-year price and total returns 2021-2025	S&P Dow Jones Indices, Market Attributes: U.S. Equities (Dec 2021, Jan 2023, Jan 2024, Dec 2024, Dec 2025); FRED series SP500
S10	Market value of all U.S.-listed domestic companies, 2020-2025	SIFMA Capital Markets Fact Book 2026, Table 28 (data from the World Federation of Exchanges); matches World Bank series CM.MKT.LCAP.CD
S11	Who owns U.S. corporate stock; public vs closely held; S vs C corporations	Federal Reserve, Financial Accounts of the United States (Z.1), Table F.51.1.s Corporate equities, release with 2026:Q2 data
S12	Wealth and stock holdings by wealth group	Federal Reserve, Distributional Financial Accounts, release of Sept 2026 (data through 2026:Q2)
S13	U.S. energy-related CO2 emissions, 2020-2025	U.S. Energy Information Administration, Monthly Energy Review, Sept 2026, Table 11.1
S14	Carbon fee path: $43/ton in 2021 rising 5% a year above inflation; border adjustment; ~$2,000 per family of four	Climate Leadership Council, "Exceeding Paris" (updated Sept 2019) and "Our Plan" page
S15	How much the fee cuts emissions, 2021-2025	Resources for the Future, Issue Brief 18-07 (revised Mar 2019), Table 1a, reprinted in CLC "Exceeding Paris"
S16	Who pays a carbon fee by income; middle-family numbers	U.S. Treasury, Office of Tax Analysis, Working Paper 115 (Horowitz et al., Jan 2017), Tables 5 and 6
S17	U.S. internet advertising revenue 2020-2025; search share	IAB/PwC Internet Advertising Revenue Reports (FY2020-FY2024 reports; FY2025 release Apr 16, 2026)
S19	Walmart: pre-tax income, employees, median pay	Walmart 10-K filings FY2022-FY2026 and proxy statements (SEC EDGAR, CIK 104169)
S22	Company market values at each Dec 31	Stock Analysis market cap history (exact trading dates; Nasdaq Data Link); cross-checked with CompaniesMarketCap

ID	What it supports	Source
S23	Domestic corporate profits before tax	BEA, NIPA Table 1.14, corporate business profits before tax (without IVA and CCAdj), via FRED A446RC1A027NBEA
S24	U.S. GDP, 2021-2025	BEA via FRED (GDPA), after the Sept 30, 2026 revision
S37	Published 10-year stock return forecasts	Vanguard (as of Jun 30, 2026: 4.2%-6.2% nominal); J.P. Morgan LTCMA 2027 (U.S. large cap 7.5%); BlackRock CMA file (Aug 2026: 8.97%; AI productivity boom scenario 15.9%); Research Affiliates (Dec 2025: 3.1%); GMO (Sept 2026: -7.3% real, 7 years)
S38	Population growth ahead	CBO, The Demographic Outlook: 2026 to 2056 (Jan 7, 2026)
S43	Stock valuation level (Shiller CAPE)	Robert Shiller's data; Multpl
S50	The earlier model of the plan	Everybody_Owns_a_Piece_Model.xlsx (an earlier spreadsheet of the plan, built with ChatGPT)
S51	How much S&P 500 revenue comes from abroad	S&P DJI "S&P 500 2018: Global Sales" (Aug 2019); S&P DJI Indexology blog (Feb 4, 2025); Goldman Sachs data via Visual Capitalist (2023); Apollo chart of FactSet data (Jan 2025)